

Hypothesis Tests and Confidence Intervals in Multiple Regression

SW Chapter 7

ECON3500: Econometrics and Applications

Spring 2026

Learning Objectives

Construct and interpret **confidence intervals** for individual coefficients in multiple regression

Test hypotheses about **one coefficient** (review from Ch5)

Test hypotheses about **one restriction involving multiple coefficients**

Construct and interpret **joint hypothesis tests** (the F-test)

Explain why testing coefficients **one at a time** can be misleading

Three Types of Hypothesis Tests

A Taxonomy of Tests

Type	Null Hypothesis	Example	Statistic
1. One restriction, one coefficient	$H_0 : \beta_j = \beta_{j,0}$	$H_0 : \beta_1 = 0$	t
2. One restriction, multiple coefficients	$H_0 : \beta_j = \beta_m$	$H_0 : \beta_1 = \beta_2$	t
3. Multiple restrictions (joint test)	$H_0 : \beta_j = \beta_{j,0}, \beta_m = \beta_{m,0}, \dots$	$H_0 : \beta_1 = \beta_2 = 0$	F

Confidence Intervals in Multiple Regression

Confidence Intervals for Individual Coefficients

In simple regression (Ch5), we constructed CIs for β_1 . The same logic extends to **any coefficient** in a multiple regression:

$$\hat{\beta}_j \pm c_{\alpha/2} \cdot SE(\hat{\beta}_j)$$

where $c_{\alpha/2}$ is the critical value from the standard normal (large n):

Confidence Level	
90%	1.645
95%	1.96
99%	2.58

i Interpretation

In 95% of possible samples that might be drawn, the confidence interval will contain the true value of β_j ."

Side Note: What CIs Are and Are Not

CIs are NOT:

The probability that the parameter is in the interval

Our confidence we have the right answer

A statement about the parameter after we observe it

CIs ARE:

- Constructed using a procedure that works 95% of the time
- Frequentist: the interval contains the true parameter in 95% of repeated samples
- Before sampling, our procedure has 95% coverage

In sum: - **Before sampling:** Our procedure has 95% coverage probability - **After sampling:** The interval either contains the parameter or it doesn't

*There's a 95% chance that **a** CI contains the true parameter, but not that **the** estimated CI contains the true parameter.*

CIs in Multiple Regression: What Changes?

Compared to simple regression:

The **formula is the same**: $\hat{\beta}_j \pm c \cdot SE(\hat{\beta}_j)$

The **standard errors change** because they now account for correlations among regressors

Key insight: Adding control variables can either **increase or decrease** the SE of $\hat{\beta}_j$:

Reduces SE if the added variable explains variation in Y (reduces σ_u^2)

Increases SE if the added variable is correlated with X_j (multicollinearity)

1. One Restriction, One Coefficient: $\beta_j = \beta_{j,0}$

Select your significance level ($\alpha = 0.01$)

State your null hypothesis:

- $H_0 : \beta_j = \beta_{j,0}$
- $H_a : \beta_j \neq \beta_{j,0}$

Compute the t-statistic:

$$t = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}$$

Compare the t-statistic to your critical value (2.58) and reject null if $|t| > c$

(Optional) Construct your confidence interval:

$$\left(\hat{\beta}_j - 2.58 \cdot SE(\hat{\beta}_j), \hat{\beta}_j + 2.58 \cdot SE(\hat{\beta}_j) \right)$$

2. One Restriction, Two Coefficients: $\beta_j = \beta_m$

Select your significance level ($\alpha = 0.01$)

State your null hypothesis: $H_0 : \beta_1 = \beta_2$ vs $H_a : \beta_1 \neq \beta_2$

Transform your regression:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + u_i$$

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{1i} - \beta_2 x_{1i} + \beta_2 x_{2i} + u_i$$

$$y_i = \beta_0 + (\beta_1 - \beta_2)x_{1i} + \beta_2(x_{1i} + x_{2i}) + u_i$$

$$y_i = \beta_0 + \gamma_1 x_{1i} + \beta_2 w_i + u_i$$

and instead test $H_0 : \gamma_1 = 0$ vs. $H_1 : \gamma_1 \neq 0$

Repeat remaining steps

3. Multiple Restrictions (Under Homoskedasticity)

There are lots of variants, but this is the one we will compute by hand

Select your significance level ($\alpha = 0.01$)

State your null hypothesis: $H_0 : \beta_1 = \beta_2 = \beta_3$ vs any part not true

Estimate model with variables you are testing (unrestricted) and without (restricted)

Calculate F -statistic

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k_{ur} - 1)} = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k_{ur} - 1)} \sim F_{q, n - k_{ur} - 1}$$

Compare the F -statistic to your critical value from a $F_{q, n - k_{ur} - 1}$ distribution and reject null if $F > c$ (usually from $F_{q, \infty}$ distribution)

Extended example: The STAR Experiment

Angrist, Lang, and Oreopoulos (2009)

Incentives And Services For College Achievement: Evidence From A Randomized Trial

By JOSHUA ANGRIST, DANIEL LANG, AND PHILIP OREOPOULOS*

This paper reports on an experimental evaluation of strategies designed to improve academic performance among college freshmen. One treatment group was offered academic support services. Another was offered financial incentives for good grades. A third group combined both interventions. Service use was highest for women and for subjects in the combined group. The combined treatment also raised the grades and improved the academic standing of women. These differentials persisted through the end of second year, though incentives were given in the first year only. This suggests study skills among some treated women increased. In contrast, the program had no effect on men. (JEL I21, I28)

We're going to think about predictors of year 1 college GPA.

The STAR Experiment: Context

Setting: A satellite campus of a large Canadian university (University of Toronto at Scarborough)

Research question: Can academic support services and financial incentives improve first-year academic performance?

Three treatment groups (randomly assigned):

SSP (Student Support Program): peer advising and facilitated study groups

SFP (Student Fellowship Program): merit-based scholarships for meeting GPA targets

SFSP: combined SSP + SFP treatment

Key findings: The combined program (SFSP) improved grades for women, particularly those with weaker high school backgrounds. Support services alone or financial incentives alone had limited effects.

Our focus today: We'll use the **control variables** from this study — gender, mother tongue, and high school quartile — to practice hypothesis testing. This is about the *predictors* of Year 1 GPA, not the treatment effects.

Recall: The Dummy Variable Trap (Ch6)

When including a categorical variable with m categories, include $m - 1$ **dummy variables** and leave one as the **reference group**.

In our example:

Mother tongue has 3 categories: English, French, Other

We include `mt_french` and `mt_other`; **English is the excluded reference group**

Coefficients on `mt_french` and `mt_other` are interpreted *relative to English speakers*

Quick Check

If we included all three dummies plus an intercept, what would happen?

The Base Regression

```
regress GPA_year1 female mt_french mt_other hs_q2 hs_q3, robust
```

```
. regress GPA_year1 female mt_french mt_other hs_q2 hs_q3
```

Source	SS	df	MS	Number of obs	=	1,374
Model	186.639466	5	37.3278932	F(5, 1368)	=	57.06
Residual	894.929455	1,368	.654188198	Prob > F	=	0.0000
				R-squared	=	0.1726
				Adj R-squared	=	0.1695
Total	1081.56892	1,373	.787741385	Root MSE	=	.80882

GPA_year1	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
female	-.1296654	.0442666	-2.93	0.003	-.2165032	-.0428276
mt_french	-.4658419	.3079241	-1.51	0.131	-1.069896	.1382127
mt_other	-.0755215	.04796	-1.57	0.116	-.1696047	.0185616
hs_q2	.3743258	.0545463	6.86	0.000	.2673224	.4813292
hs_q3	.8857912	.0532672	16.63	0.000	.781297	.9902854
_cons	1.484259	.0467597	31.74	0.000	1.392531	1.575987

Controls for gender, mother tongue, and HS quartile (no top quartile!)

Choice of Excluded Group Only Affects Interpretation

English mother tongue excluded:

$$\widehat{GPA_{year1}} = 1.484 - 0.130 \cdot female - 0.466 \cdot mt_{french} - 0.076 \cdot mt_{other} + 0.374$$

“Other” mother tongue excluded:

$$\widehat{GPA_{year1}} = 1.409 - 0.130 \cdot female - 0.076 \cdot mt_{english} - 0.390 \cdot mt_{french} + 0.37$$

Choice of Excluded Group Only Affects Interpretation

For simplicity, assume $female = 0$, $hs_{q2} = 0$, and $hs_{q3} = 0$

English mother tongue excluded:

$$\widehat{GPA_{year1}} = 1.484 - 0.130 \cdot female - 0.466 \cdot mt_{french} - 0.0755 \cdot mt_{other} + 0.37$$

$$E[GPA_{year1} | mt_{english} = 1] = 1.484$$

$$E[GPA_{year1} | mt_{french} = 1] = 1.484 - 0.466 = 1.018$$

$$E[GPA_{year1} | mt_{other} = 1] = 1.4843 - 0.0755 = 1.409$$

Choice of Excluded Group Only Affects Interpretation (cont.)

For simplicity, assume $female = 0$, $hs_{q2} = 0$, and $hs_{q3} = 0$

“Other” mother tongue excluded:

$$\widehat{GPA_{year1}} = 1.409 - 0.130 \cdot female + 0.0755 \cdot mt_{english} - 0.390 \cdot mt_{french} + 0.3$$

$$E[GPA_{year1} | mt_{english} = 1] = 1.4087 + 0.0755 = 1.484$$

$$E[GPA_{year1} | mt_{french} = 1] = 1.4087 - 0.3903 = 1.018$$

$$E[GPA_{year1} | mt_{other} = 1] = 1.4087 = 1.409$$

Questions We Could Want to Answer

```
. regress GPA_year1 female mt_french mt_other hs_q2 hs_q3
```

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GPA_year1	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
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mt_other	-.0755215	.04796	-1.57	0.116	-.1696047	.0185616
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Do French speakers have the same GPA as English speakers? → **Type 1**: test $\beta_{french} = 0$

Do “Other” speakers have the same GPA as English speakers? → **Type 1**: test $\beta_{other} = 0$

Do French speakers have the same GPA as “Other” speakers? → **Type 2**

Are there *any* differences in GPA by mother tongue? → **Type 3**

Testing Equality of Two Coefficients

Do French speakers have the same GPA as “Other” speakers?

Population model:

$$GPA_i = \beta_0 + \beta_1 \cdot mt_{french} + \beta_2 \cdot mt_{other} + \beta_3 \cdot hS_{q2} + \beta_4 \cdot hS_{q3} + u_i$$

$$H_0 : \beta_1 = \beta_2 \quad \text{vs.} \quad H_a : \beta_1 \neq \beta_2$$

Equivalently: $H_0 : \beta_1 - \beta_2 = 0$

Problem: We can't directly test this with a standard t -test on a single coefficient.

Solution: **Transform the regression.**

The Transformation Trick

Start with:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + u_i$$

Add and subtract $\beta_2 x_{1i}$:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{1i} - \beta_2 x_{1i} + \beta_2 x_{2i} + \cdots + u_i$$

Rearrange:

$$y_i = \beta_0 + \underbrace{(\beta_1 + \beta_2)}_{\gamma_1} x_{1i} + \underbrace{\beta_2 (x_{1i} + x_{2i})}_{w_i} + \cdots + u_i$$

Now test: $H_0 : \gamma_1 = 0$ — a standard **Type 1** t -test!

In Stata: The Easy Way

After running the regression, Stata's `test` command does this automatically:

```
regress GPA_year1 female mt_french mt_other hs_q2 hs_q3  
test mt_french = mt_other
```

Result:

$$F(1, 1368) = 1.59 \quad p = 0.2073$$

We **cannot reject** the null that French and “Other” speakers have the same GPA ($p = 0.21$).

F vs. t with One Restriction

With a single restriction, $F = t^2$, so the F -test and two-sided t -test are equivalent. Stata reports F from the `test` command.

Why Not Just Test One at a Time?

Are there *any* differences in GPA by mother tongue?

Tempting approach: Just look at the individual t -statistics!

$\hat{\beta}_{french}: t = -1.51, p = 0.131 \rightarrow$ not significant

$\hat{\beta}_{other}: t = -1.57, p = 0.116 \rightarrow$ not significant

Conclusion: Mother tongue doesn't matter?

This Is Wrong!

Testing coefficients one at a time and concluding “none are significant, so the group doesn't matter” is a **logical error**.

Each individual test has, say, a 5% chance of Type I error. But when you run **multiple tests**, the probability that *at least one* falsely rejects grows quickly.

The Multiple Testing Problem

Suppose you test q hypotheses, each at the 5% level, and all nulls are true.

Probability of **not** rejecting any single test: 0.95

Probability of not rejecting **any** of q independent tests: 0.95^q

Probability of **at least one** false rejection: $1 - 0.95^q$

Number of tests	P(at least one false rejection)
1	5.0%
2	9.8%
5	22.6%
10	40.1%

We need a test that evaluates **all restrictions simultaneously**: the **F-test**.

Joint Hypothesis Test: Setup

Population model:

$$GPA_i = \beta_0 + \beta_1 \cdot mt_{french} + \beta_2 \cdot mt_{other} + \beta_3 \cdot hS_{q2} + \beta_4 \cdot hS_{q3} + u_i$$

$$H_0 : \beta_1 = 0 \text{ and } \beta_2 = 0 \quad \text{vs.} \quad H_a : \beta_1 \neq 0 \text{ and/or } \beta_2 \neq 0$$

This is a test of whether the mother tongue coefficients are **jointly significant**.

Equivalently: can we **exclude** `mt_french` and `mt_other` from the model?

Key Terms

Exclusion Restriction

A test of whether certain covariates can be excluded from the population model.

Unrestricted model: the model with **more** covariates

- “Unrestricted” because those coefficients are free to be zero or anything else

Restricted model: the model with **fewer** covariates

- “Restricted” because we are **imposing** that those coefficients equal zero

The Unrestricted Model

The full model with all variables — including the ones we're testing:

```
. regress GPA_year1 female mt_french mt_other hs_q2 hs_q3
```

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Note the highlighted values: $SSR_{ur} = 894.93$ and the coefficients on `mt_french` and `mt_other`.

The Restricted Model

The model **without** mother tongue — imposing $\beta_1 = \beta_2 = 0$:

```
. regress GPA_year1 female hs_q2 hs_q3
```

Source	SS	df	MS	Number of obs	=	1,374
Model	183.662596	3	61.2208653	F(3, 1370)	=	93.41
Residual	897.906326	1,370	.655406077	Prob > F	=	0.0000
				R-squared	=	0.1698
				Adj R-squared	=	0.1680
Total	1081.56892	1,373	.787741385	Root MSE	=	.80957

GPA_year1	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
female	-.1333114	.0442706	-3.01	0.003	-.2201568	-.0464659
hs_q2	.3686178	.0545086	6.76	0.000	.2616884	.4755472
hs_q3	.8780338	.0531684	16.51	0.000	.7737334	.9823341
_cons	1.466264	.045171	32.46	0.000	1.377652	1.554875

$SSR_r = 897.91$ — the sum of squared residuals **necessarily increases** when we drop variables.

But is that increase **statistically significant**?

The F-Statistic (Homoskedastic Version)

$$F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k_{ur} - 1)} \sim F_{q, n-k_{ur}-1}$$

where:

q = number of restrictions being tested

$n - k_{ur} - 1$ = degrees of freedom in the unrestricted model

SSR_r = sum of squared residuals from the restricted model

SSR_{ur} = sum of squared residuals from the unrestricted model

Intuition: The F -statistic measures the **relative increase in SSR** when we impose the restrictions. If the null is true, this increase should be small.

Equivalent Formula Using R^2

Since $SSR = TSS \cdot (1 - R^2)$ and TSS is the same in both models:

$$F = \frac{(R_{ur}^2 - R_r^2)/q}{(1 - R_{ur}^2)/(n - k_{ur} - 1)}$$

This version is useful when regression output reports R^2 but not SSR .

The F-Distribution

The F -distribution:

Takes only **positive values** (reflecting that SSR can only increase when we drop variables)

Has two parameters: q (numerator df) and $n - k_{ur} - 1$ (denominator df)

Rejection is **one-sided**: reject when $F > c$

Choose the critical value c so that the null is falsely rejected in $\alpha\%$ of cases.

Computing the F-Test: By Hand

From the Stata output: $SSR_{ur} = 894.93$, $SSR_r = 897.91$, $q = 2$, $n = 1374$, $k_{ur} = 5$

$$F = \frac{(897.91 - 894.93)/2}{894.93/(1374 - 5 - 1)} = \frac{2.98/2}{894.93/1368} = \frac{1.49}{0.654} = 2.28$$

$$F \sim F_{2,1368} \rightarrow c_{0.10} = 2.30$$

$$P(F > 2.28) = 0.1032$$

We **cannot reject** the null at conventional significance levels.

The mother tongue variables are not **jointly significant** — we cannot conclude that mother tongue predicts GPA.

In Stata: `test` and `testparm`

After the unrestricted regression, Stata can compute the F-test directly:

```
* Test specific equality restrictions
test mt_french = mt_other = 0

* Or equivalently, test joint significance of a group
testparm mt_french mt_other
```

Both give: $F(2, 1368) = 2.28, p = 0.1032$

Stata Tip

`testparm` is convenient for testing whether a **group of variables** is jointly significant — it sets up the $H_0 : \beta_j = 0$ for each variable in the list.

Don't Forget Heteroskedasticity!

The F-statistic formula using *SSR* assumes **homoskedasticity**.

In practice, always estimate with **robust standard errors**:

```
regress GPA_year1 female mt_french mt_other hs_q2 hs_q3, robust  
test mt_french = mt_other = 0
```

Robust F-test result: $F(2, 1368) = 2.86, p = 0.0573$

⚠ Notice the Difference

Under homoskedasticity: $F = 2.28, p = 0.103$

With robust SEs: $F = 2.86, p = 0.057$

The robust version is **closer to significance**. Always use robust standard errors unless you have a specific reason not to.

Test of Overall Significance

Question: Do *any* of our explanatory variables predict y ?

Null hypothesis: All slope coefficients are zero:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_k = 0$$

Restricted model (under H_0):

$$y_i = \beta_0 + u_i$$

(Just the sample mean — no predictors at all)

Unrestricted model:

$$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \cdots + \beta_k x_{ki} + u_i$$

The F-statistic:

$$F = \frac{R^2/k}{(1 - R^2)/(n - k - 1)} \sim F_{k, n-k-1}$$

Why we care: If this F-test is not significant, your model explains essentially *none* of the variation in y — a red flag!

You've Been Seeing This All Along

```
. regress GPA_year1 female mt_french mt_other hs_q2 hs_q3
```

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The **F(5, 1368) = 57.06** and **Prob > F = 0.0000** at the top of every Stata regression output is the test of overall significance!

i Usually Overwhelmingly Rejected

If your overall F-test is *not* significant, your model has no predictive power — a red flag!

In Stata: `testparm *`

You can also compute it manually after the regression:

```
testparm *
```

$$F(5, 1368) = 57.06 \quad p = 0.0000$$

This confirms what Stata already reports at the top of the regression output.

When to Use Which Test?

t-Test vs. F-Test

Aspect	t-test	F-test
# of restrictions	One	One or more
Sided?	One or two-sided	Two-sided only
Use for	Single coefficient	Multiple coefficients

 Key Relationship

When testing a **single restriction**: $F = t^2$

The results are functionally equivalent! The F-test is a **generalization** of the two-sided *t*-test to multiple restrictions.

Recipe Card: Hypothesis Tests in Multiple Regression

i Type 1: One Restriction, One Coefficient

$$H_0 : \beta_j = \beta_{j,0} \text{ (usually } \beta_{j,0} = 0 \text{)}$$

$$\text{Statistic: } t = \frac{\hat{\beta}_j - \beta_{j,0}}{SE(\hat{\beta}_j)}$$

Distribution: Standard normal (large n)

Stata: Read directly from regression output, or `test varname = value`

i Type 2: One Restriction, Multiple Coefficients

$$H_0 : \beta_j = \beta_m$$

Transform the regression so the restriction becomes $\gamma_1 = 0$

Or use Stata: `test var1 = var2`

i Type 3: Joint Hypothesis (F-Test)

$$H_0 : \beta_1 = 0, \beta_2 = 0, \dots$$

$$\text{Statistic (under homoskedasticity): } F = \frac{(SSR_r - SSR_{ur})/q}{SSR_{ur}/(n - k_{ur} - 1)}$$

Distribution: $F_{q, n - k_{ur} - 1}$

Stata: `test` or `testparm`

Aside: Homoskedasticity, F-Tests, and BLUE

We’ve seen homoskedasticity come up twice now:

Ch5: Needed for Gauss-Markov (OLS is BLUE)

Ch7 (today): The hand-computed F -statistic formula assumes it

i Quick Recap: When Does Homoskedasticity Matter?

Context	Homoskedasticity needed?
OLS point estimates ($\hat{\beta}$)	No — unbiased either way
OLS is BLUE (most efficient)	Yes — the “B” in BLUE
Homoskedastic-only SEs / F-stats	Yes — wrong if violated
Robust SEs / robust F-stats	No — valid either way

Bottom line: Use robust SEs. Then the only thing you lose when homoskedasticity fails is the guarantee that OLS is the most efficient linear estimator — and that’s usually a small

Key Takeaways

Confidence intervals in multiple regression use the same formula as simple regression, but SEs account for other regressors

Individual t -tests work for single restrictions — but **don't test one at a time** when you have a joint hypothesis

The **F-test** tests multiple restrictions simultaneously, avoiding the multiple testing problem

The **overall F-test** is reported at the top of every regression — it tests whether all regressors are jointly significant

Always use **robust standard errors** for heteroskedasticity-robust inference

Knowledge Check

If each of 5 individual t -tests fails to reject at the 5% level, can you conclude the variables are jointly insignificant? Why or why not?



Tip

Answer: No! The joint F-test could still reject.

Individual insignificance does not imply joint insignificance — this is precisely why we need the F-test.

Appendix: GiveDirectly Kenya Study

GiveDirectly Context: Why Cash Transfers?

The Challenge

700+ million people live on less than \$2/day

Traditional Approaches

In-kind aid (food, blankets, etc.)

Conditional cash transfers (CCTs): “We give you money IF you send your kids to school / get vaccinated”

Paternalistic: Government decides what the poor “need”

Unconditional Cash Transfers (UCTs) - Direct, no strings attached - Trusts recipients to make their own decisions - Respects agency and autonomy

Why This Matters

Economic Argument

If markets work, cash is most efficient → recipients choose optimally

But markets often fail in poor areas:

Limited local supply (firms don't invest)

Prices may spike if aggregate demand increases (inflation)

Spillovers: Money spent in one household affects neighbors

Egger, Haushofer, Miguel, Niehaus, Walker (2022)

An alternative application of hypothesis testing using field experimental data on unconditional cash transfers.

Paper: “General Equilibrium Effects of Cash Transfers: Experimental Evidence from Kenya”

Published in *Econometrica* 90(6):2603–2643 | **2024 Frisch Medal Winner** 🏆

Research Question: What are the direct and general equilibrium effects of unconditional cash transfers?

The Study Design

Study Context

653 villages in rural Kenya (Siaya County)

10,500+ poor households

Cash transfer: ~\$1,000 USD (~87,000 KES) per eligible household

Fiscal shock: 15% of local GDP

Randomization Strategy

Village-level: Treatment vs. control villages

Household-level: Eligible vs. ineligible within treatment villages

- Eligible (thatched roof means test) → receive transfer
- Ineligible (better housing) → NO transfer but exposed to spillovers

Main Findings

Direct Effects (on eligible recipients)

Consumption increased by 1,200-1,800 KES/month (~\$12-18/month)

Assets increased substantially

Income effects from local multipliers

Spillover Effects (on ineligible neighbors)

Consumption increased by 500-1,200 KES/month (positive spillovers!)

Local firms benefited from increased demand

Minimal price inflation (no evidence of inflation eroding gains)

Local Fiscal Multiplier: 2.4

For every dollar transferred, local economic activity increased by \$2.40

Suggests powerful local demand effects and limited import leakage

Bottom Line: Cash transfers helped both direct recipients AND their neighbors through general equilibrium effects

Regression Framework

Key Variables

`treatment` = 1 if household in treatment village (any status)

`eligible` = 1 if household eligible for transfer (means test)

`ineligible` = 1 if household in treatment village but ineligible

Outcome: Monthly household consumption (PPP-adjusted, KES)

Controls: Female household head, log household size, age of head (in decades)

Interaction Terms Used

`treatment × eligible` captures direct effect

`treatment × ineligible` captures spillover effect

Can test equality of these effects (Type 2 test)